Answer 03

Here, we need to compare the Favourite Algorithm and Decision Tree Induction using 10 fold cross validation.

Difference = Decision Tree Error Rate − Favourite Alg Rate

Fold	Difference
1	0.41
2	−0.04
3	0.99
4	−0.35
5	−0.14
6	0.78
7	0.46
8	−0.56
9	0.00
10	−0.03

Avg Diff = $\frac{1.52}{10} = 0.152$

Standard Dev = 0.494

T-value = $0.152 / (0.494/\sqrt{10})$

= 0.97.

Degree of freedom = 10 − 1

= 09.

* Using the t-table for 9 degree of freedom, a t-value of 0.97 gives approximately 82% one sided confidence.

So, we can be approximately 82% confident that the Favourite Algorithm perform better then the decision Tree Induction algorithm.